

HW12

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Exercise 12.1

某种商品一周的需要量是一个随机变量, 其密度函数为

$$p_1(t) = \begin{cases} te^{-t}, & t > 0, \\ 0, & t \leq 0. \end{cases}$$

设各周的需要量是相互独立的, 试求:

1. 两周需要量的密度函数 $p_2(x)$;
2. 三周需要量的密度函数 $p_3(x)$.

Proof. 设 X_1, X_2, X_3 分别为第一, 二和三周分别的需要量, 设 Y_2, Y_3 为前两周和前三周的需要量. 则

$$Y_2 = X_1 + X_2, \quad Y_3 = Y_2 + X_3$$

$$p_{X_1}(t) = p_{X_2}(t) = p_{X_3}(t) = p_1(t), \quad p_{Y_2}(t) = p_2(t), \quad p_{Y_3}(t) = p_3(t)$$

1.

$$\begin{aligned} p_{Y_2}(t) &= p_{X_1} * p_{X_2}(t) = p_1 * p_1(t) \\ &= \int_{\mathbb{R}} p_1(t-x) p_1(x) dx \\ &= \int_{\mathbb{R}} \chi_{\{t-x>0\}}(t-x) e^{x-t} \chi_{\{x>0\}} x e^{-x} dx \\ &= \int_{\mathbb{R}} \chi_{\{t>0\}} \chi_{\{x<t\}} \chi_{\{x>0\}} (t-x) x e^{x-t} e^{-x} dx \\ &= \chi_{\{t>0\}} \int_0^t x(t-x) e^{-t} dx \\ &= \chi_{\{t>0\}} \frac{1}{6} t^3 e^{-t} \end{aligned}$$

故

$$p_2(x) = \begin{cases} \frac{1}{6}x^3 e^{-x}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

2.

$$\begin{aligned} p_{Y_3}(t) &= p_{X_1} * p_{Y_2}(t) = p_1 * p_2(t) \\ &= \int_{\mathbb{R}} \chi_{\{t-x>0\}}(t-x) e^{x-t} \chi_{\{x>0\}} \frac{1}{6} x^3 e^{-x} dx \\ &= \int_{\mathbb{R}} \chi_{\{x<t\}} \chi_{\{x>0\}} \frac{1}{6} x^3 (t-x) e^{-t} dx \\ &= \chi_{\{t>0\}} \int_0^t \frac{1}{6} x^3 (t-x) e^{-t} dx \\ &= \chi_{\{t>0\}} \left(\frac{1}{24} t^5 e^{-t} - \frac{1}{30} t^5 e^{-t} \right) \\ &= \chi_{\{t>0\}} \frac{1}{120} t^5 e^{-t} \end{aligned}$$

因此

$$p_3(x) = \begin{cases} \frac{1}{120} x^5 e^{-x}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

□

Exercise 12.2

设 X_1, X_2, X_3 为相互独立的随机变量, 且都服从 $(0, 1)$ 上的均匀分布, 求三者中最大者大于其他两者之和的概率.

Proof. 由对称性, 所求概率等于

$$\begin{aligned} 3P(X_3 > X_1 + X_2) &= 3 \iint_{[0,1]^3} \chi_{\{x_3 > x_1 + x_2\}} dx_1 dx_2 dx_3 \\ &= 3 \int_0^1 dx_1 \int_0^{1-x_1} dx_2 \int_{x_1+x_2}^1 dx_3 \\ &= 3 \times \frac{1}{6} = \frac{1}{2} \end{aligned}$$

因此所需概率为 $\frac{1}{2}$

□

Exercise 12.3

设随机变量 X_1 与 X_2 相互独立同分布, 其密度函数为

$$p(x) = \begin{cases} 2x, & 0 < x < 1, \\ 0, & \text{其它.} \end{cases} \quad (12.1)$$

试求 $Z = \max\{X_1, X_2\} - \min\{X_1, X_2\}$ 的分布.

Solution. 易见 Z 的取值非负, 任取 $t \geq 0$, 由对称性我们有

$$P(Z > t) = 2P(X_1 - X_2 > t)$$

由独立性,

$$p_{(X_1, X_2)}(s, t) = p_{X_1}(s) p_{X_2}(t) = \chi_{\{0 < s < 1\}} \chi_{\{0 < t < 1\}} 4st$$

因此

$$\begin{aligned} P(X_1 - X_2 > t) &= \iint_{(0,1)^2} \chi_{\{x_1 > x_2 + t\}} 4x_1 x_2 \, dx_1 \, dx_2 \\ &= \chi_{\{0 < t < 1\}} \int_0^{1-t} \int_{x_2+t}^1 4x_1 x_2 \, dx_1 \, dx_2 \\ &= \chi_{\{0 < t < 1\}} \int_0^{1-t} 2x_2 (1 - (x_2 + t)^2) \, dx_2 \\ &= \chi_{\{0 < t < 1\}} \frac{1}{6} (1-t)^3 (3+t) \end{aligned}$$

因此

$$P(Z \leq t) = 1 - P(Z > t) = 1 - \chi_{\{0 < t < 1\}} \frac{1}{3} (1-t)^3 (3+t)$$

$$p_Z(t) = \frac{4}{3} (1-t)^2 (2+t) \chi_{\{0 < t < 1\}}$$

◇

Exercise 12.4

设二维随机变量 (X, Y) 服从圆心在原点的单位圆内的均匀分布, 求极坐标

$$R = \sqrt{X^2 + Y^2}, \quad \Theta = \arctan(Y/X)$$

的联合密度函数.

Solution. 令

$$D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$$

做变量替换

$$\varphi(r, \theta) = (x(r, \theta), y(r, \theta)) = (r \cos \theta, r \sin \theta)$$

则

$$\varphi^{-1}(D) = \{(r, \theta) \in [0, 1) \times [0, 2\pi)\}$$

由于

$$f_{(X,Y)}(x, y) = \chi_D \frac{1}{\pi}$$

则由变量替换公式,

$$f_{(R,\Theta)}(r, \theta) = \chi_{\varphi^{-1}(D)} f_{X,Y}(x(r, \theta), y(r, \theta)) \cdot |J|$$

其中

$$J = \frac{\partial(x, y)}{\partial(r, \theta)} = \det \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix} = r$$

因此

$$f_{(R,\Theta)}(r, \theta) = \chi_{\varphi^{-1}(D)} \frac{|r|}{\pi} = \chi_{\{0 \leq r < 1\}} \chi_{\{0 \leq \theta < 2\pi\}} \frac{r}{\pi}$$

◇

Exercise 12.5

设随机变量 X 与 Y 独立同分布, 且都服从标准正态分布 $N(0, 1)$, 试证: $U = X^2 + Y^2$ 与 $V = X/Y$ 相互独立.

Proof. 令

$$R = \sqrt{X^2 + Y^2}, \quad \Theta = \arg(X + iY)$$

做变量替换 $\varphi: \mathbb{R}_{\geq 0} \times [0, 2\pi) \rightarrow \mathbb{R}^2$

$$\varphi(r, \theta) = (x(r, \theta), y(r, \theta)) = (r \cos \theta, r \sin \theta)$$

令 $D = \mathbb{R}_{\geq 0} \times [0, 2\pi)$ 则

$$\begin{aligned} f_{R,\Theta}(r, \theta) &= \chi_D f_{X,Y}(r \cos \theta, r \sin \theta) \cdot |J| \\ &= \chi_D f_X(r \cos \theta) f_Y(r \sin \theta) r \\ &= \chi_D \frac{1}{2\pi} e^{-\frac{r^2}{2}} r \\ &= \left(r e^{-\frac{r^2}{2}} \chi_{\{r \geq 0\}} \right) \cdot \left(\frac{1}{2\pi} \chi_{\{\theta \in [0, 2\pi)\}} \right) \end{aligned}$$

因此 $f_{R,\Theta}$ 是可分离变量的, R, Θ 相互独立. 注意到

$$U = R^2, \quad V = \cot \Theta$$

由于独立随机变量的可测函数也是独立的, 得到 U, V 独立. □

Exercise 12.6

设随机变量 X 与 Y 相互独立, 且 $X \sim Ga(\alpha_1, \lambda)$, $Y \sim Ga(\alpha_2, \lambda)$. 试证: $U = X + Y$ 与 $V = X/(X + Y)$ 相互独立, 且 $V \sim Be(\alpha_1, \alpha_2)$.

Proof. 由 Gamma 分布的可加性,

$$U = X + Y \sim Ga(\alpha_1 + \alpha_2, \lambda)$$

考虑变量替换 $u = x + y, v = \frac{x}{x+y}, (x, y) \in (0, \infty)^2$ 则

$$x = uv, \quad y = u - uv,$$

$$(u, v) \in D = \{(u, v) \in \mathbb{R}^2 : 0 < u < \infty, 0 < v < 1\}$$

我们有

$$f_{U,V}(u, v) = \chi_D f_{X,Y}(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right|$$

其中

$$\frac{\partial(x, y)}{\partial(u, v)} = \det \begin{pmatrix} v & u \\ 1-v & -u \end{pmatrix} = -uv - (u - uv) = -u$$

$$\begin{aligned} f_{X,Y}(x, y) &= f_X(x) f_Y(y) = \frac{\lambda^{\alpha_1}}{\Gamma(\alpha_1)} x^{\alpha_1-1} e^{-\lambda x} \frac{\lambda^{\alpha_2}}{\Gamma(\alpha_2)} y^{\alpha_2-1} e^{-\lambda y} \\ &= \frac{\lambda^{\alpha_1+\alpha_2}}{\Gamma(\alpha_1) \Gamma(\alpha_2)} x^{\alpha_1-1} y^{\alpha_2-1} e^{-\lambda(x+y)} \end{aligned}$$

故

$$\begin{aligned}
 f_{U,V}(u, v) &= \chi_D \frac{\lambda^{\alpha_1 + \alpha_2}}{\Gamma(\alpha_1) \Gamma(\alpha_2)} u^{\alpha_1 - 1} v^{\alpha_1 - 1} u^{\alpha_2 - 1} (1 - v)^{\alpha_2 - 1} e^{-\lambda u} \\
 &= \left(\chi_{\{0 < u < \infty\}} \frac{\lambda^{\alpha_1 + \alpha_2}}{\Gamma(\alpha_1 + \alpha_2)} u^{\alpha_1 + \alpha_2 - 1} e^{-\lambda u} \right) \left(\chi_{\{0 < v < 1\}} \frac{\Gamma(\alpha_1 + \alpha_2)}{\Gamma(\alpha_1) \Gamma(\alpha_2)} v^{\alpha_1 - 1} (1 - v)^{\alpha_2 - 1} \right) \\
 &= f_U(u) \left(\chi_{\{0 < v < 1\}} \frac{\Gamma(\alpha_1 + \alpha_2)}{\Gamma(\alpha_1) \Gamma(\alpha_2)} v^{\alpha_1 - 1} (1 - v)^{\alpha_2 - 1} \right)
 \end{aligned}$$

是关于 u, v 分离变量的, 故 U, V 独立, 那么

$$f_{U,V}(u, v) = f_U(u) f_V(v)$$

得到

$$f_V(v) = \left(\chi_{\{0 < v < 1\}} \frac{\Gamma(\alpha_1 + \alpha_2)}{\Gamma(\alpha_1) \Gamma(\alpha_2)} v^{\alpha_1 - 1} (1 - v)^{\alpha_2 - 1} \right)$$

这就表明 V 服从 $Be(\alpha_1, \alpha_2)$ □

Exercise 12.7

设随机变量 U_1 与 U_2 相互独立, 且都服从 $(0, 1)$ 上的均匀分布, 试证明:

$$Z_1 = -2 \ln U_1 \sim \text{Exp}(1/2), \quad Z_2 = 2\pi U_2 \sim U(0, 2\pi); \quad (12.2)$$

$$X = \sqrt{Z_1} \cos Z_2 \text{ 和 } Y = \sqrt{Z_1} \sin Z_2 \text{ 是相互独立的标准正态随机变量.} \quad (12.3)$$

Proof.

$$f_{U_1}(u_1) = \chi_{\{0 < u_1 < 1\}}, \quad f_{U_2}(u_2) = \chi_{\{0 < u_2 < 1\}}$$

考虑变量替换 $z_1 = -2 \ln u_1$, 则 $u_1 = \exp\left(-\frac{z_1}{2}\right)$

$$\begin{aligned}
 f_{Z_1}(z_1) &= f_{U_1}(u_1(z_1)) \left| \frac{\partial u_1}{\partial z_1} \right| = \chi_{\{0 < \exp(-\frac{z_1}{2}) < 1\}} \frac{1}{2} \exp\left(-\frac{z_1}{2}\right) \\
 &= \chi_{\{z_1 > 0\}} \frac{1}{2} \exp\left(-\frac{z_1}{2}\right)
 \end{aligned}$$

这表明 $Z_1 \sim \text{Exp}\left(\frac{1}{2}\right)$.

考虑变量替换 $z_2 = 2\pi u_2$, 则 $u_2 = \frac{z_2}{2\pi}$, 我们有

$$\begin{aligned} f_{Z_2}(z_2) &= f_{U_2}(u_2(z_2)) \left| \frac{\partial u_2}{\partial z_2} \right| \\ &= \chi_{\{0 < \frac{z_2}{2\pi} < 1\}} \frac{1}{2\pi} = \chi_{\{0 < z_2 < 2\pi\}} \frac{1}{2\pi} \end{aligned}$$

这表明 $Z_2 \sim U(0, 2\pi)$. 考虑变量替换

$$x = \sqrt{z_1} \cos z_2, \quad y = \sqrt{z_1} \sin z_2$$

则

$$z_1 = x^2 + y^2, \quad z_2 = \arg(x + iy)$$

$$\begin{aligned} f_{X,Y}(x, y) &= \chi_{\{z_1 > 0\}} \chi_{\{0 < z_2 < 2\pi\}} f_{Z_1, Z_2}(z_1(x, y), z_2(x, y)) \left| \frac{\partial(z_1, z_2)}{\partial(x, y)} \right| \\ &= f_{Z_1}(x^2 + y^2) f_{Z_2}(\arg(x + iy)) \left| \det \begin{pmatrix} \frac{2x}{x^2 + y^2} & \frac{2y}{x^2 + y^2} \\ -\frac{y}{x^2 + y^2} & \frac{x}{x^2 + y^2} \end{pmatrix} \right| \\ &= \frac{1}{2} \exp\left(-\frac{x^2 + y^2}{2}\right) \frac{1}{2\pi} 2 \\ &= \left(\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) \right) \left(\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{y^2}{2}\right) \right) \end{aligned}$$

故 X, Y 是相互独立的标准正态随机变量. □

Exercise 12.8

求掷 n 颗骰子出现点数之和的数学期望与方差。

Proof. 设 $X_1, \dots, X_n$ 分别表示 n 颗骰子的点数. 则由期望的线性性,

$$E(X_1 + \dots + X_n) = E(X_1) + \dots + E(X_n)$$

$$E(X_i) = \sum_{k=1}^6 P(X_i = k) k = \frac{7}{2}, \quad i = 1, \dots, n$$

故

$$E(X_1 + \dots + X_n) = \frac{7n}{2}$$

由于 $X_1, \dots, X_n$ 之间相互独立, 我们有

$$\text{Var}(X_1 + \dots + X_n) = \text{Var}(X_1) + \dots + \text{Var}(X_n)$$

而

$$\text{Var}(X_i) = \sum_{k=1}^6 P(X_i = k) \left(k - \frac{7}{2}\right)^2 = \frac{35}{12}$$

故

$$\text{Var}(X_1 + \dots + X_n) = \frac{35n}{12}$$

□

Exercise 12.9

从数字 $0, 1, \dots, n$ 中任取两个不同的数字, 求这两个数字之差的绝对值的数学期望.

Proof. 设 X 和 Y 分别为任取的两个不同的数字. 样本空间的大小为 $A_{n+1}^2 = n(n+1)$. 令 $Z = |X - Y|$, Z 的可能取值为 $k = 1, 2, \dots, n$. 对于固定的 k , 满足 $|X - Y| = k$ 的数对 (X, Y) 有:

$$(k, 0), (k+1, 1), \dots, (n, n-k) \quad \text{以及} \quad (0, k), (1, k+1), \dots, (n-k, n)$$

共 $2(n-k+1)$ 种情况. 故

$$P(Z = k) = \frac{2(n-k+1)}{n(n+1)}, \quad k = 1, 2, \dots, n$$

$$\begin{aligned} E(Z) &= \sum_{k=1}^n k \cdot P(Z = k) \\ &= \sum_{k=1}^n k \frac{2(n-k+1)}{n(n+1)} \\ &= \frac{2}{n(n+1)} \sum_{k=1}^n ((n+1)k - k^2) \\ &= \frac{2}{n(n+1)} \left((n+1) \sum_{k=1}^n k - \sum_{k=1}^n k^2 \right) \end{aligned}$$

代入求和公式 $\sum_{k=1}^n k = \frac{n(n+1)}{2}$ 和 $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$, 得到

$$\begin{aligned}
 E(Z) &= \frac{2}{n(n+1)} \left((n+1) \frac{n(n+1)}{2} - \frac{n(n+1)(2n+1)}{6} \right) \\
 &= \frac{2}{n(n+1)} \cdot \frac{n(n+1)}{2} \left((n+1) - \frac{2n+1}{3} \right) \\
 &= (n+1) - \frac{2n+1}{3} \\
 &= \frac{3n+3-2n-1}{3} \\
 &= \frac{n+2}{3}
 \end{aligned}$$

□

Exercise 12.10

设在区间 $(0, 1)$ 上随机地取 n 个点, 求相距最远的两点间的距离的数学期望.

Proof. 设 $X_1, \dots, X_n$ 是这 n 个点, 定义

$$X_{(1)} = \min \{X_1, \dots, X_n\}, \quad X_{(n)} = \max \{X_1, \dots, X_n\}$$

则相距最远的两点间的距离 $D = X_{(n)} - X_{(1)}$. 我们有

$$F(X_{(n)}) = F_{X_1}(x) \cdots F_{X_n}(x) = \chi_{\{0 < x < 1\}} x^n$$

$$f_{X_{(n)}}(x) = \chi_{\{0 < x < 1\}} n x^{n-1}$$

$$1 - F(X_{(1)}) = (1 - F_{X_1}(x)) \cdots (1 - F_{X_n}(x)) = (1 - F_{X_1}(x))^n$$

我们有

$$\begin{aligned}
 f_{X_{(1)}}(x) &= \frac{d}{dx} (1 - (1 - F_{X_1}(x)))^n = n(1 - F_{X_1}(x))^{n-1} f_{X_1}(x) \\
 &= n(1 - \chi_{\{0 < x < 1\}} x + \chi_{\{x \geq 1\}} 1)^{n-1} \chi_{\{0 < x < 1\}} \\
 &= \chi_{\{0 < x < 1\}} n(1 - x)^{n-1}
 \end{aligned}$$

于是

$$E(X_{(n)}) = \int_0^1 n x^n dx = \frac{n}{n+1}$$

$$\begin{aligned}
E(X_{(1)}) &= \int_0^1 nx(1-x)^{n-1} dx \\
&= \int_0^1 n(1-x)x^{n-1} dx \\
&= \int_0^1 nx^{n-1} dx - \int_0^1 nx^n dx \\
&= 1 - \frac{n}{n+1} = \frac{1}{n+1}
\end{aligned}$$

故

$$E(D) = E(X_{(n)}) - E(X_{(1)}) = \frac{n-1}{n+1}$$

□

Exercise 12.11

盒中有 n 个不同的球, 其上分别写有数字 $1, 2, \dots, n$, 每次随机抽出一个, 记下其号码, 放回去再抽. 直到抽到有两个不同的数字为止, 求平均抽球次数.

Solution. 记 N 为停止时的总抽球次数. 设 X 为第一次抽球的点数. 此后每次抽球都可以看作是一次独立的伯努利试验, “成功”(即抽到与 x 不同的球)的概率为 $p = \frac{n-1}{n}$.

令 Y 表示第一次之后直到停止所需的额外抽球次数, 则 Y 服从参数为 p 的几何分布, 其期望为 $E[Y] = \frac{1}{p}$. 由期望的线性性质, 总次数的期望为:

$$\begin{aligned}
E[N] &= 1 + E[Y] \\
&= 1 + \frac{1}{p} \\
&= 1 + \frac{n}{n-1} \\
&= \frac{2n-1}{n-1}
\end{aligned}$$

◇

Exercise 12.12

设随机变量 (X, Y) 的联合分布列为

	$Y = 0$	$Y = 1$
$X = 0$	0.1	0.15
$X = 1$	0.25	0.2
$X = 2$	0.15	0.15

试求 $Z = \sin\left(\frac{\pi}{2}(X + Y)\right)$ 的数学期望.

Proof.

$$\begin{aligned}
 E[Z] &= \sum_{x=0}^2 \sum_{y=0}^1 \sin\left(\frac{\pi}{2}(x+y)\right) P(X=x, Y=y) \\
 &= \sin(0)P(0,0) + \sin\left(\frac{\pi}{2}\right)[P(0,1) + P(1,0)] \\
 &\quad + \sin(\pi)[P(1,1) + P(2,0)] + \sin\left(\frac{3\pi}{2}\right)P(2,1) \\
 &= 0 + 1 \cdot (0.15 + 0.25) + 0 + (-1) \cdot 0.15 \\
 &= 0.4 - 0.15 \\
 &= 0.25.
 \end{aligned}$$

□

Exercise 12.13

随机变量 (X, Y) 服从以点 $(0, 1), (1, 0), (1, 1)$ 为顶点的三角形区域上的均匀分布, 试求 $E(X + Y)$ 和 $\text{Var}(X + Y)$.

Proof. 区域 D 的面积为 $1/2$, 故概率密度函数为 $f(x, y) = 2$. 积分区域可表示为 $0 \leq x \leq 1, 1 - x \leq y \leq 1$.

$$E(X+Y) = \int_0^1 dx \int_{1-x}^1 (x+y) \cdot 2 dy = \int_0^1 [(x+y)^2]_{1-x}^1 dx = \int_0^1 ((x+1)^2 - 1) dx = \frac{4}{3}$$

$$E((X + Y)^2) = \int_0^1 dx \int_{1-x}^1 (x + y)^2 \cdot 2 dy = \frac{2}{3} \int_0^1 ((x + 1)^3 - 1) dx = \frac{11}{6}$$

$$\text{Var}(X + Y) = E((X + Y)^2) - (E(X + Y))^2 = \frac{11}{6} - \frac{16}{9} = \frac{1}{18}$$

□

Exercise 12.14

设 X, Y 为 $(0, 1)$ 上独立的均匀随机变量, 试证:

$$E(|X - Y|^\alpha) = \frac{2}{(\alpha + 1)(\alpha + 2)}, \alpha > 0.$$

Proof.

$$\begin{aligned} f_{X,Y}(x, y) &= \chi_{\{0 < x < 1\}} \chi_{\{0 < y < 1\}} \\ E(|X - Y|^\alpha) &= \int_0^1 \int_0^1 |x - y|^\alpha dx dy \\ &= \int_0^1 \int_y^1 (x - y)^\alpha dx dy + \int_0^1 \int_0^y (y - x)^\alpha dx dy \\ &= \int_0^1 \frac{1}{\alpha + 1} (1 - y)^{\alpha+1} dy + \int_0^1 \frac{1}{\alpha + 1} y^{\alpha+1} dy \\ &= \frac{1}{(\alpha + 1)(\alpha + 2)} + \frac{1}{(\alpha + 1)(\alpha + 2)} \\ &= \frac{2}{(\alpha + 1)(\alpha + 2)} \end{aligned}$$

□

Exercise 12.15

设随机变量 X 与 Y 独立同分布, 且 $E(X) = \mu$, $\text{Var}(X) = \sigma^2$. 试求 $E(X - Y)^2$.

Proof.

$$E((X - Y)^2) = E(X^2) - 2E(XY) + E(Y^2) = 2(\sigma^2 + \mu^2) - 2E(XY)$$

$$\begin{aligned}
 E(XY) &= \iint xy f_X(x) f_Y(y) dx dy \\
 &= \left(\int x f_X(x) dx \right) \left(\int y f_Y(y) dy \right) \\
 &= E(X) E(Y) = \mu^2
 \end{aligned}$$

故

$$E((X - Y)^2) = 2\sigma^2$$

□

Exercise 12.16

设随机变量 (X, Y) 的联合密度函数为

$$p(x, y) = \begin{cases} \frac{x(1 + 3y^2)}{4}, & 0 < x < 2, 0 < y < 1, \\ 0, & \text{其他.} \end{cases}$$

试求 $E(Y/X)$.

Solution.

$$\begin{aligned}
 E\left(\frac{Y}{X}\right) &= \int_0^2 \int_0^1 \frac{y}{x} p(x, y) dy dx \\
 &= \int_0^2 \int_0^1 \frac{1}{4} y (1 + 3y^2) dy dx \\
 &= \int_0^2 \frac{1}{8} + \frac{3}{16} dx \\
 &= \frac{5}{8}
 \end{aligned}$$

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Exercise 12.17

系统由 n 个部件组成. 记 X_i 为第 i 个部件能持续工作的时间, 如果 $X_1, X_2, \dots, X_n$ 独立同分布, 且 $X_i \sim \text{Exp}(\lambda)$, 试在以下情况下求系统持续工作的平均时间:

- (1) 如果有一个部件停止工作, 系统就不工作了;
- (2) 如果至少有一个部件在工作, 系统就工作.

Exercise 12.18

设 X, Y 独立同分布, 都服从标准正态分布 $N(0, 1)$, 求 $E(\max\{X, Y\})$.

Proof. 由于 $X, Y \sim N(0, 1)$,

$$\begin{aligned}
 f_{\max\{X, Y\}}(t) &= 2\phi(t)\Phi(t) = \frac{1}{\pi} \int_{-\infty}^t e^{-\frac{x^2+t^2}{2}} dx \\
 E(\max\{X, Y\}) &= \frac{1}{\pi} \int_{-\infty}^{\infty} \int_{-\infty}^t t e^{-\frac{x^2+t^2}{2}} dx dt \\
 &= \frac{1}{\pi} \int_{-\infty}^{\infty} e^{-\frac{x^2}{2}} \left(\int_x^{\infty} t e^{-\frac{t^2}{2}} dt \right) dx \\
 &= \frac{1}{\pi} \int_{-\infty}^{\infty} e^{-\frac{x^2}{2}} \cdot e^{-\frac{x^2}{2}} dx \\
 &= \frac{1}{\pi} \int_{-\infty}^{\infty} e^{-x^2} dx \\
 &= \frac{1}{\pi} \cdot \sqrt{\pi} = \frac{1}{\sqrt{\pi}}
 \end{aligned}$$

□

Exercise 12.19

设随机变量 U 服从 $(-2, 2)$ 上的均匀分布, 定义 X 和 Y 如下:

$$X = \begin{cases} -1, & \text{若 } U < -1, \\ 1, & \text{若 } U \geq -1, \end{cases} \quad Y = \begin{cases} -1, & \text{若 } U < 1, \\ 1, & \text{若 } U \geq 1. \end{cases}$$

试求 $\text{Var}(X + Y)$.

Proof.

$$X = \chi_{\{-1 < U < 2\}} - \chi_{\{-2 < U < -1\}}, \quad Y = \chi_{\{1 < U < 2\}} - \chi_{\{-2 < U < 1\}}$$

$$E(X) = P(-1 < U < 2) - P(-2 < U < -1) = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}$$

$$E(Y) = P(1 < U < 2) - P(-2 < U < 1) = \frac{1}{4} - \frac{3}{4} = -\frac{1}{2}$$

$$E(X + Y) = E(X) + E(Y) = 0$$

$$X = \chi_{\{-1 < U < 1\}} + \chi_{\{1 < U < 2\}} - \chi_{\{-2 < U < -1\}}, \quad a.e.$$

$$Y = \chi_{\{1 < U < 2\}} - \chi_{\{-2 < U < -1\}} - \chi_{\{-1 < U < 1\}}, \quad a.e.$$

故

$$X + Y = 2\chi_{\{1 < U < 2\}} - 2\chi_{\{-2 < U < -1\}}, \quad a.e.$$

$$\begin{aligned} \text{Var}(X + Y) &= E((X + Y)^2) \\ &= \int (X + Y)^2 dP \\ &= \int 4\chi_{\{1 < U < 2\}} + 4\chi_{\{-2 < U < -1\}} dP \\ &= 4P(\{1 < U < 2\}) + 4P(\{-2 < U < -1\}) \\ &= 2 \end{aligned}$$

□

Exercise 12.20

设随机变量 X 与 Y 独立, 都服从正态分布 $N(a, \sigma^2)$, 试证:

$$E(\max\{X, Y\}) = a + \frac{\sigma}{\sqrt{\pi}}.$$

Proof. 利用恒等式 $\max\{X, Y\} = \frac{X+Y+|X-Y|}{2}$, 由期望的线性性质可得:

$$E(\max\{X, Y\}) = \frac{1}{2}E(X) + \frac{1}{2}E(Y) + \frac{1}{2}E(|X - Y|).$$

已知 $X, Y \sim N(a, \sigma^2)$, 故 $E(X) = E(Y) = a$.

令 $Z = X - Y$. 由于 X 与 Y 相互独立且服从正态分布, 其线性组合 Z 仍服从正态分布.

$$E(Z) = E(X) - E(Y) = a - a = 0,$$

$$\text{Var}(Z) = \text{Var}(X) + \text{Var}(Y) = \sigma^2 + \sigma^2 = 2\sigma^2.$$

即 $Z \sim N(0, 2\sigma^2)$. 其概率密度函数为 $f_Z(z) = \frac{1}{\sqrt{2\pi} \cdot \sqrt{2}\sigma} e^{-\frac{z^2}{4\sigma^2}} = \frac{1}{2\sqrt{\pi}\sigma} e^{-\frac{z^2}{4\sigma^2}}.$

$$\begin{aligned} E(|Z|) &= \int_{-\infty}^{\infty} |z| f_Z(z) dz \\ &= 2 \int_0^{\infty} z \cdot \frac{1}{2\sqrt{\pi}\sigma} e^{-\frac{z^2}{4\sigma^2}} dz \\ &= \frac{1}{\sqrt{\pi}\sigma} \int_0^{\infty} z e^{-\frac{z^2}{4\sigma^2}} dz. \end{aligned}$$

作变量代换, 令 $u = \frac{z^2}{4\sigma^2}$, 则 $du = \frac{z}{2\sigma^2} dz$, 即 $z dz = 2\sigma^2 du$. 积分变为:

$$\int_0^{\infty} z e^{-\frac{z^2}{4\sigma^2}} dz = \int_0^{\infty} e^{-u} \cdot 2\sigma^2 du = 2\sigma^2.$$

代回 $E(|Z|)$ 的表达式:

$$E(|X - Y|) = E(|Z|) = \frac{1}{\sqrt{\pi}\sigma} \cdot 2\sigma^2 = \frac{2\sigma}{\sqrt{\pi}}.$$

故

$$E(\max\{X, Y\}) = a + \frac{1}{2}E(|X - Y|) = a + \frac{1}{2} \cdot \frac{2\sigma}{\sqrt{\pi}} = a + \frac{\sigma}{\sqrt{\pi}}.$$

□